Chapter 6
. Interconnection Networks and Clusters

1. Interconnection Networks Media

   There is a hierarchy of media to interconnect computers that varies in cost, performance, and reliability. Network media have another figure of merit, the maximum distance between nodes. This section covers three popular examples, and Figure 8.11 illustrates them.

Category 5 Unsheilded Twisted pair ("Cat5"):

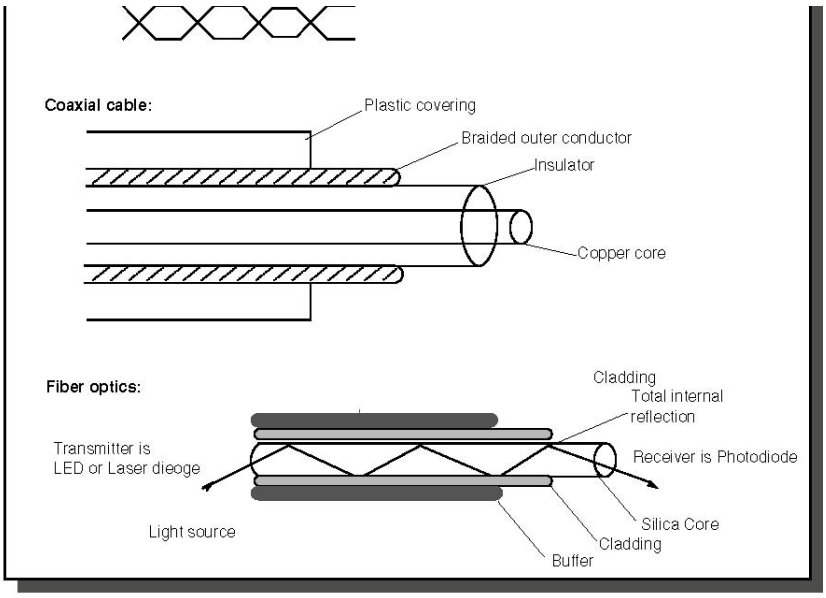


FIGURE 8.11 Three network media. (From a presentation by David Culler of U.C. Berkeley.)

The frst medium is *twisted pairs* of copper wires. These are two insulated wires, each about 1 mm thick. They are twisted together to reduce electrical interference, since two parallel lines form an antenna but a twisted pair does not. As they can transfer a few megabits per second over several kilometers without amplification, twisted pair were the mainstay of the telephone system. Telephone companies bundled together (and sheathed) many pairs coming into a building. Twisted pairs can also offer tens of megabits per second of bandwidth over shorter distances, making them plausible for LANs.

The original telephone-line quality was called Level 1. Level 3 was good enough for 10 Mbits/second Ethernet. The desire for even greater bandwidth lead to the Lev-el 5 or Category 5, which is sufficient for 100 Mbits/second Ethernet. By limiting the length to 100 meters, "Cat5" wiring can be used for 1000 Mbits/second Ethernet links today. It uses the RJ-45 connector, which is similar to the connector found on telephone lines.

Coaxial cable was deployed by cable television companies to deliver a higher rate over a few kilometers. To offer high bandwidth and good noise immunity, insulating material surrounds a single stiff copper wire, and then cylindrical conductor surrounds the insulator, often woven as a braided mesh. A 50-ohm baseband coaxial cable delivers 10 megabits per second over a kilometer.

The third transmission media is Fiber optics which transmits digital data as pulses of light. A fiber optic network has three components:

1 the transmission medium, a fiber optic cable;

2 the light source, an LED or laser diode;

3 the light detector, a photodiode.

First, cladding surrounds the glass fiber core to confine the light. A buffer then surrounds the cladding to protect the core and cladding. Note that unlike twisted pairs or coax, fibers are one-way, or *simplex,* media. A two-way, or *full duplex,* connection between two nodes requires two fibers.

Since light bends or refracts at interfaces, it can slowly spread as it travels down the cable unless the diameter of the cable is limited to one wavelength of light; then it transfers in a straight line. Thus, fiber optic cables are of two forms:

1. *Multimode fiber*—It uses inexpensive LEDs as a light source. It is typically much larger than the wavelength of light: typically 62.5 microns in diameter vs. the 1.3-micron wavelength of infrared light. Since it is wider it has more dispersion problems, where some wave frequencies have different propaga-tion velocities. The LEDs and dispersion limit it to up to a few hundred meters at 1000 Mbits/second or a few kilometers at 100 Mbits /second. It is older and less expensive than single mode fiber.

2. *Single-mode fiber*—This “single-wavelength” fiber (typically 8 to 9 microns in diameter) requires more expensive laser diodes for light sources and currently transmits gigabits per second for hundreds of kilometers, making it the medium of choice for telephone companies. The loss of signal strength as it passes through a medium, called *attenuation*, limits the length of the fiber.

To achieve even more bandwidth from a fiber, *wavelength division multiplexing (WDM)* sends different streams simultaneously on the same fiber using different wavelengths of light, and then demultiplexes the different wavelengths at the receiver. In 2001, WDM can deliver a combined 40 Gbits/second using about 8 wavelengths, with plans to go to 80 wavelengths and deliver 400 Gbits/second.

Practical Issues for Commercial Interconnection Networks :

There are practical issues in addition to the technical issues described so far that are important considerations for some interconnection networks: connectivity, standardization, and fault tolerance.

Connectivity

The number of machines that communication affects the complexity of the net-work and its protocols. The protocols must target the largest size of the network, and handle the types of anomalous events that occur. Hundreds of machines communicating are a much easier than millions.

Connecting the Network to the Computer

Computers have a hierarchy of buses with different cost/performance. For example, a personal computer in 2001 has a memory bus, a PCI bus for fast I/O de-vices, and an USB bus for slow I/O devices. I/O buses follow open standards and have less stringent electrical requirements. Memory buses, on the other hand, provide higher bandwidth and lower latency than I/O buses. Where to connect the network to the machine depends on the performance

goals, all LANs and WANs plug into the I/O bus.

The location of the network connection significantly affects the software interface to the network as well as the hardware. A memory bus is more likely to be cache-coherent than an I/O bus and therefore more likely to avoid these extra cache flishes. DMA is the best way to send large messages. Whether to use DMA to send small messages depends on the efficiency of the interface to the DMA. The DMA interface is usually memory-mapped, and so each interaction is typically at the speed of main memory rather than of a cache access.

Standardization: Cross-Company Interoperability

LANs and WANs use standards and interoperate effectively. WANs involve many types of companies and must connect to many brands of computers, so it is difficult to imagine a proprietary WAN ever being successful. The ubiquitous nature of the Ethernet shows the popularity of standards for LANs as well as WANs, and it seems unlikely that many customers would tie the viability of their LAN to the stability of a single company.

Message Failure Tolerance

The communication system must have mechanisms for retransmission of a message in case of failure. Often it is handled in higher layers of the software protocol at the end points, requiring retransmission at the source. Given the long time of flight for WANs, often they can retransmit from hop to hop rather relying only on retransmission from the source.

Node Failure Tolerance

The second practical issue refers to whether or not the interconnection relies on all the nodes being operational in order for the interconnection to work properly. Since software failures are generally much more frequent than hardware failures, the question is whether a software crash on a single node can prevent the rest of the nodes from communicating.

Clearly, WANs would be useless if they demanded that thousands of computers spread

across a continent be continuously available, and so they all tolerate the failures of individual nodes. LANs connect dozens to hundreds of computers together, and again it would be impractical to require that no computer ever fail. All successful LANs normally survive node failures.

Clusters

There are many mainframe applications—such as databases, file servers, Web servers, simulations, and multiprogramming/batch processing—amenable to running on more loosely coupled machines than the cache-coherent NUMA machines. These applications often need to be highly available, requiring some form of fault tolerance and repairability. Such applications—plus the similarity of the multiprocessor nodes to desktop computers and the emergence of high-bandwidth, switch-based local area networks—lead to clusters of off-the-shelf, whole computers for large-scale processing.

Performance Challenges of Clusters

One drawback is that clusters are usually connected using the I/O bus of the computer, whereas multiprocessors are usually connected on the memory bus of the computer. The memory bus has higher bandwidth and much lower latency, allowing multiprocessors to drive the network link at higher speed and to have fewer conflicts with I/O traffic on I/O-intensive applications. This connection point also means that clusters generally use software-based communication while multiprocessors use hardware for communication. However, it makes connections non-standard and hence more expensive.

A second weakness is the division of memory: a cluster of N machines has N independent memories and N copies of the operating system, but a shared address multiprocessor allows a single program to use almost all the memory in the computer. Thus, a sequential program in a cluster has 1/Nth the memory available compared to a sequential program in a shared memory multiprocessor. Interestingly, the drop in DRAM prices has made memory costs so low that this multi-processor advantage is much less important in 2001 than it was in 1995. The primary issue in 2001 is whether the maximum memory per cluster node is

sufficient for the application.

Dependability and Scalability Advantage of Clusters

The weakness of separate memories for program size turns out to be a strength in system availability and expansibility. Since a cluster consists of independent computers are connected through a local area network, it is much easier to replace a machine without bringing down the system in a cluster than in an shared memory multiprocessor. Clusters are constructed from whole computers and independent, scalable networks, this isolation also makes it easier to expand the system without bringing down the application that runs on top of the cluster. High availability and rapid, incremental extensibility make clusters attractive to service providers for the World Wide Web.

Pros and Cons of Cost of Clusters

One drawback of clusters has been that the cost of ownership. Administering a cluster of *N* machines is close to the cost of administering *N* independent machines, while the cost of administering a shared address space multiprocessor with *N* processors is close to the cost of administering a single, big machine.

Another difference between the two tends to be the price for equivalent computing power for large-scale machines. Since large-scale multiprocessors have small volumes, the extra development costs of large machines must be amortized over few systems, resulting in higher cost to the customer. Originally, the partitioning of memory into separate modules in each node was a significant disadvantage to clusters, as division means memory is used less efficiently than on a shared address computer. The incredible drop in price of memory has mitigated this weakness, dramatically changed the trade-offs in favor of clusters.

Popularity of Clusters

Low cost, scaling and fault isolation proved a perfect match to the companies providing services over the Internet since the mid 1990s. Internet applications such as search engines and email servers are amenable to more loosely coupled computers, as the parallelism consists of millions of independent tasks. Hence, companies like Amazon, AOL Google, Hotmail,

Inktomi, WebTV, and Yahoo rely on clusters of PCs or workstations to provide services used by millions of people every day.

Clusters are growing in popularity in the scientific computing market as well. Figure 8.30 shows the mix of architecture styles between 1993 and 2000 for the top 500 fastest scientific computers. One attraction is that individual scientists can afford to construct clusters themselves, allowing them to dedicate their cluster to their problem. Shared supercomputers are placed on monthly allocation of CPU time, so its plausible for a scientist to get more work done from a private cluster than from a shared supercomputer. It is also relatively easy for the scientist to scale his computing over time as he gets more money for computing.

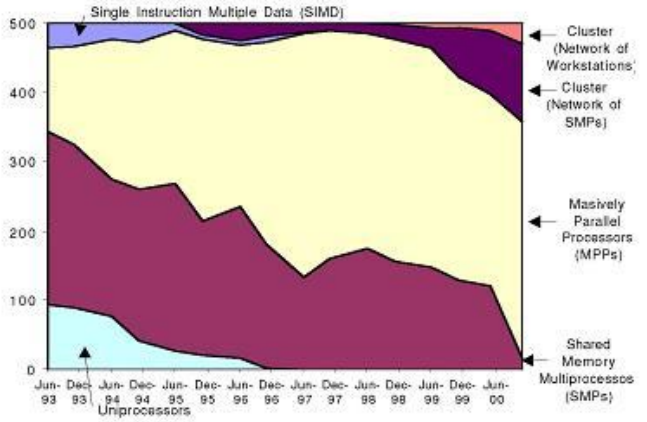


FIGURE 8.30 Plot of Top 500 supercomputer sites between 1993 and 2000. Note that clusters of various kinds grew from 2% to almost 30% in the last three years, while uniprocessors and SMPs have almost disappeared. In fact, most of the MPPs in the list look are similar to clusters. In 2001, the top 500 collectively has a performance of about 100 Teraflops [Bell 2001]. Performance is measured as speed of running Linpack, which solves a dense system of linear equations. This list at www.top500.org is updated twice a year.

Designing a Cluster:

Consider a system with about 32 processors, 32 GB of DRAM, and 32 or 64 disks. Figure 8.33 lists the components we use to construct the cluster, including their prices.

IBM model name	xSeries 300	xSeries 330	xSeries 370
Maximum number processors per box	1	2	8
Pentium III Processor Clock Rate (MHz)	1000	1000	700
L2 Cache (KB)	256	256	1,024
Price of base computer with 1 Processor	$1,759	$1,939	$14,614

Price per extra Processor	n.a.	$799	$1,779
Price per 256 MB SDRAM DIMM	$159	$269	$369
Price per 512 MB SDRAM DIMM	$549	$749	$1,069
Price per 1024 MB SDRAM DIMM	n.a.	$1,689	$2,369
IBM 36.4 GB 10K RPM Ultra160 SCSI	$579	$639	$639
IBM 73.4 GB 10K RPM Ultra160 SCSI	n.a.	$1,299	$1,299
PCI slots: 32bit,33 MHz / 64bit,33 MHz / 64bit,66 MHz	1 / 0 / 0	0 / 2 / 0	0 / 8 / 4
Rack space (VME Rack Units)	1	1	8
Power Supply	200 W	200 W	3 x 750 W
Emulex cLAN-1000 Host Adapter (1 Gbit)	$795	$795	$795
Emulex cLAN5000 8-port switch	$6,280	$6,280	$6,280
Emulex cLAN5000 Rack space (R.U.)	1	1	1
Emulex cLAN5300 30-port switch	$15,995	$15,995	$15,995
Emulex cLAN5300 Rack space (R.U.)	2	2	2
Emulex cLAN-1000 10-meter cable	$135	$135	$135
Extra PCI Ultra160 SCSI Adapter	$299	$299	$299
EXP300 Storage Enclosure (up to 14 disks)	$3,179	$3,179	$3,179
EXP300 Rack space (VME Rack Units)	3	3	3
Ultra2 SCSI 4-meter cable	$105	$105	$105
Standard 19-in Rack (44 VME Rack Units)	$1795	$1795	$1795

Figure 8.33 confirms some of the philosophical points of the prior section. Note that difference in cost and speed processor is in the smaller systems versus the larger multiprocessor. In addition, the price per DRAM DIMM goes up with the size of the computers.

The higher price of the DRAM is harder too explain based on cost. For example, all include ECC. The uni-processor uses 133 MHz SDRAM and the 2-way and 8-way both use registered

DIMM modules (RDIMM) SDRAM. There might a slightly higher cost for the buffered DRAM between the uni-processor and 2-way boxes, but it is hard to explain increasing price 1.5 times for the 8-way SMP vs. the 2-way SMP. In fact, the 8-way SDRAM operates at just 100 MHz. Pre-sumably, customers willing to pay a premium for processors for an 8-way SMP are also willing to pay more for memory.

Reasons for higher price matters little to the designer of a cluster. The task is to minimize cost for a given performance target. To motivate this section, here is an overview of the four examples

1 *Cost of Cluster Hardware Alternatives with Local Disk*: The first example compares the cost of building from a uniprocessor, a 2-way SMP, and an 8-way SMP. In this example, the disks are directly attached to the computers in the cluster.

2 *Cost of Cluster Hardware Alternatives with Disks over SAN*: The second ex-ample moves the disk storage behind a RAID controller on a Storage Area Network.

3 *Cost of Cluster Options that is more realistic*: The third example includes the cost of software, the cost of space, some maintenance costs, and operator costs.

4 *Cost and Performance of a Cluster for Transaction Processing*: This final example describes a similar cluster tailored by IBM to run the TPC-C bench-mark. This example has more memory and many more disks to achieve a high TPC-C result, and at the time of this writing, it the 13th fastest TPC-C system. In fact, the machine with the fastest TPC-C is just a replicated version of this cluster with a bigger LAN switch.

First Example: Cost of Cluster Hardware Alternatives with Local Disk

This first example looks only at hardware cost of the three alternatives using the IBM pricing information.

Second Example: Using a SAN for disks.

The previous example uses disks local to the computer. Although this can reduce costs and space, the problem for the operator is that

1) there is no protection against a single disk failure, and

2) there is state in each computer that must be managed separately.

Hence, the system is down on a disk failures until the operator arrives, and there is no separate visibility or access to storage. This second example centralizes the disks behind a RAID controller in each case using FC-AL as the Storage Area Network.

Third Example: Accounting for Other Costs

The first and second examples only calculated the cost of the hardware. There are two other obvious costs not included: software and the cost of a maintenance agreement for the hardware.

Fourth Example: Cost and Performance of a Cluster for Transaction Processing:

This cluster also has 32 processors, uses the same IBM computers as building blocks, and it uses the same switch to connect computers together.

Disk size: since TPC-C cares more about I/Os per second (IOPS) than disk capacity, this clusters uses many small fast disks. The use of small disks gives many more IOPS for the same capacity. These disks also rotate at 15000 RPM vs. 10000 RPM, delivering more IOPS per disk.

RAID: Since the TPC-C benchmark does not factor in human costs for running a system, there is little incentive to use a SAN. TPC-C does require a RAID protection of disks, however. IBM used a RAID product that plugs into a PCI card and provides four SCSI strings. To get higher availability and performance, each enclosure attaches to two SCSI buses.

Memory: Conventional wisdom for TPC-C is to pack as much DRAM as possible into the servers. Hence, each of the four 8-way SMPs is stuffed with the maximum of 32 GB, yielding a total of 128 GB.

Processor:This benchmark uses 900 MHz Pentium III with a 2MB L2 cache. The price is $6599 as compared to prior 8-way clusters for $1799 for the 700 MHz Pentium III with a 1 MB L2 cache

PCI slots: This cluster uses 7 of the 12 available PCI bus slots for the RAID controllers compared to 1 PCI bus slot for an external SCSI or FC-AL controller in the prior 8-way

clusters. This greater utilization follows the guideline of trying to use all resources of a large SMP

Tape Reader, Monitor, Uninterruptable Power Supply: To make the system easier to come up and to keep running for the benchmark, IBM includes one DLT tape reader, four monitors, and four UPSs

Maintenance and spares: TPC-C allows use of spares to reduce maintenance costs, which is a minimum of two spares or 10% of the items. Hence, there are two spare Ethernet switches, host adapters, and cables for TPC-C

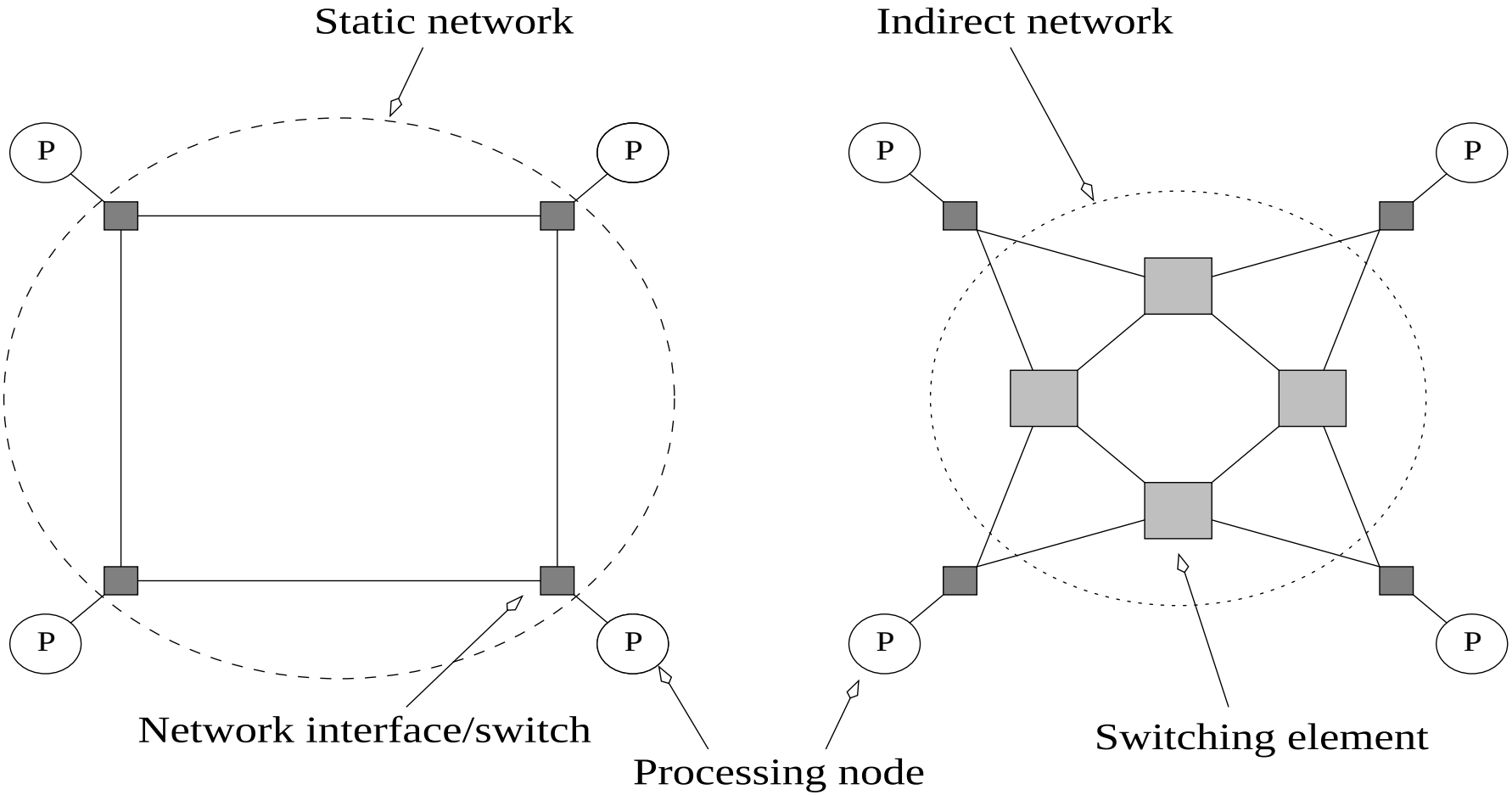
Interconnection Networks for Parallel Computers

- Interconnection networks carry data between processors and to memory.
- Interconnects are made of switches and links (wires, fiber).
- Interconnects are classified as Shared Busses, Static or Dynamic.
- Static networks consist of point-to-point communication links among processing nodes and are also referred to as *direct* networks.
- Dynamic networks are built using switches and communication links. Dynamic networks are also referred to as *indirect* networks.



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Static and Dynamic Interconnection Networks



Classification of interconnection networks: (a) a static network; and (b) a dynamic network.

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Interconnection Networks

- Switches map a fixed number of inputs to outputs.
- The total number of ports on a switch is the *degree* of the switch.
- The cost of a crossbar switch grows as the square of the degree of the switch, the peripheral hardware linearly as the degree, and the packaging costs linearly as the number of pins.



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Interconnection Networks: Network Interfaces

- Processors talk to the network via a network interface.
- The network interface may hang off the I/O bus or the memory bus.
- In a physical sense, this distinguishes a cluster from a tightly coupled multicomputer.
- The relative speeds of the I/O and memory buses impact the performance of the network.



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Network Topologies

- A variety of network topologies have been proposed and implemented.
- These topologies tradeoff performance for cost.
- Commercial machines often implement hybrids of multiple topologies for reasons of packaging, cost, and available components.



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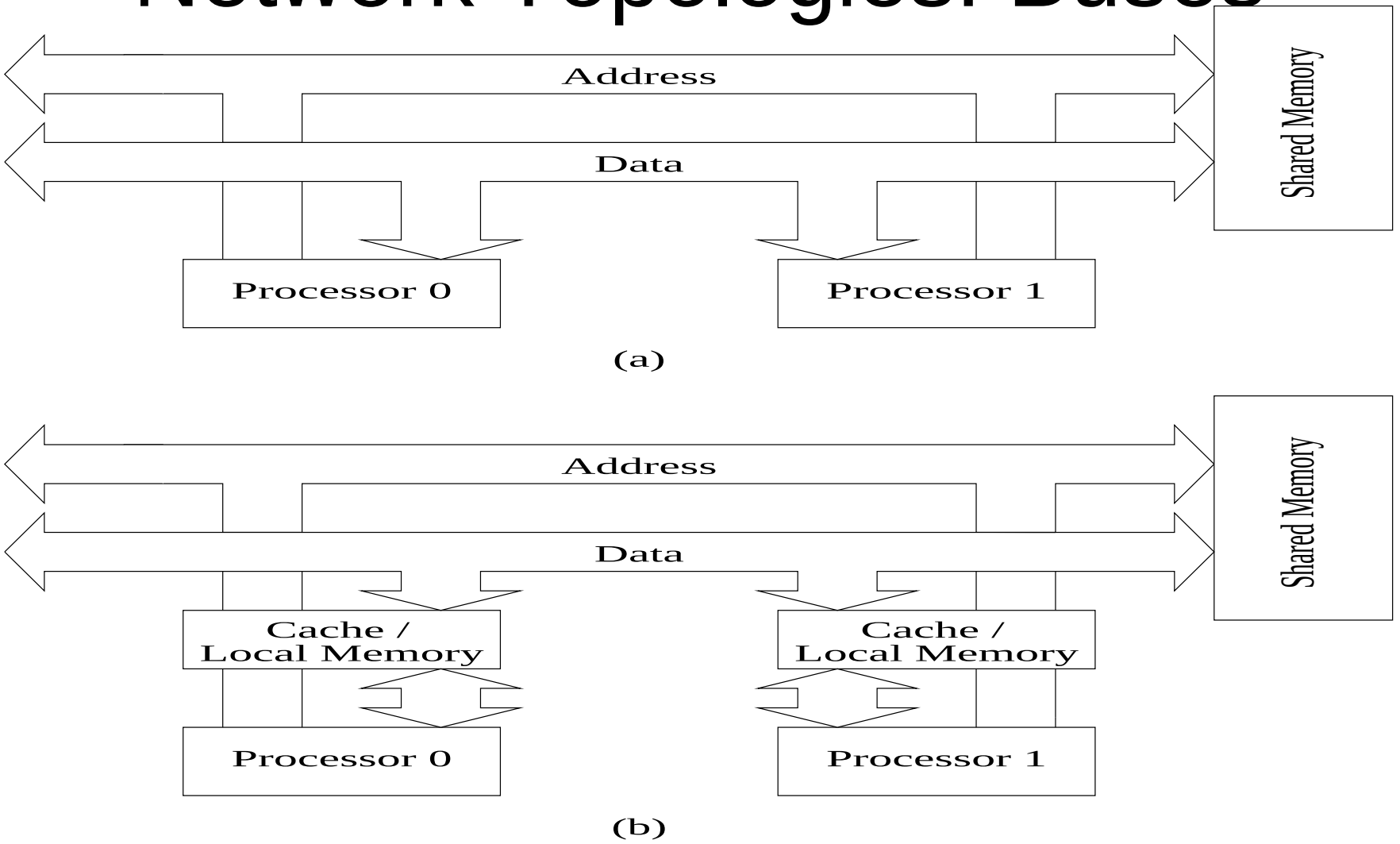
Network Topologies: Buses

- All processors access a common bus for exchanging data.
- Some of the simplest and earliest parallel machines used buses.
- The distance between any two nodes is *O(1)* in a bus. The bus also provides a convenient broadcast media.
- However, the bandwidth of the shared bus is a major bottleneck.
- Typical bus based machines are limited to dozens of nodes. Sun Enterprise servers and Intel Pentium based shared-bus multiprocessors are examples of such architectures.



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Network Topologies: Buses



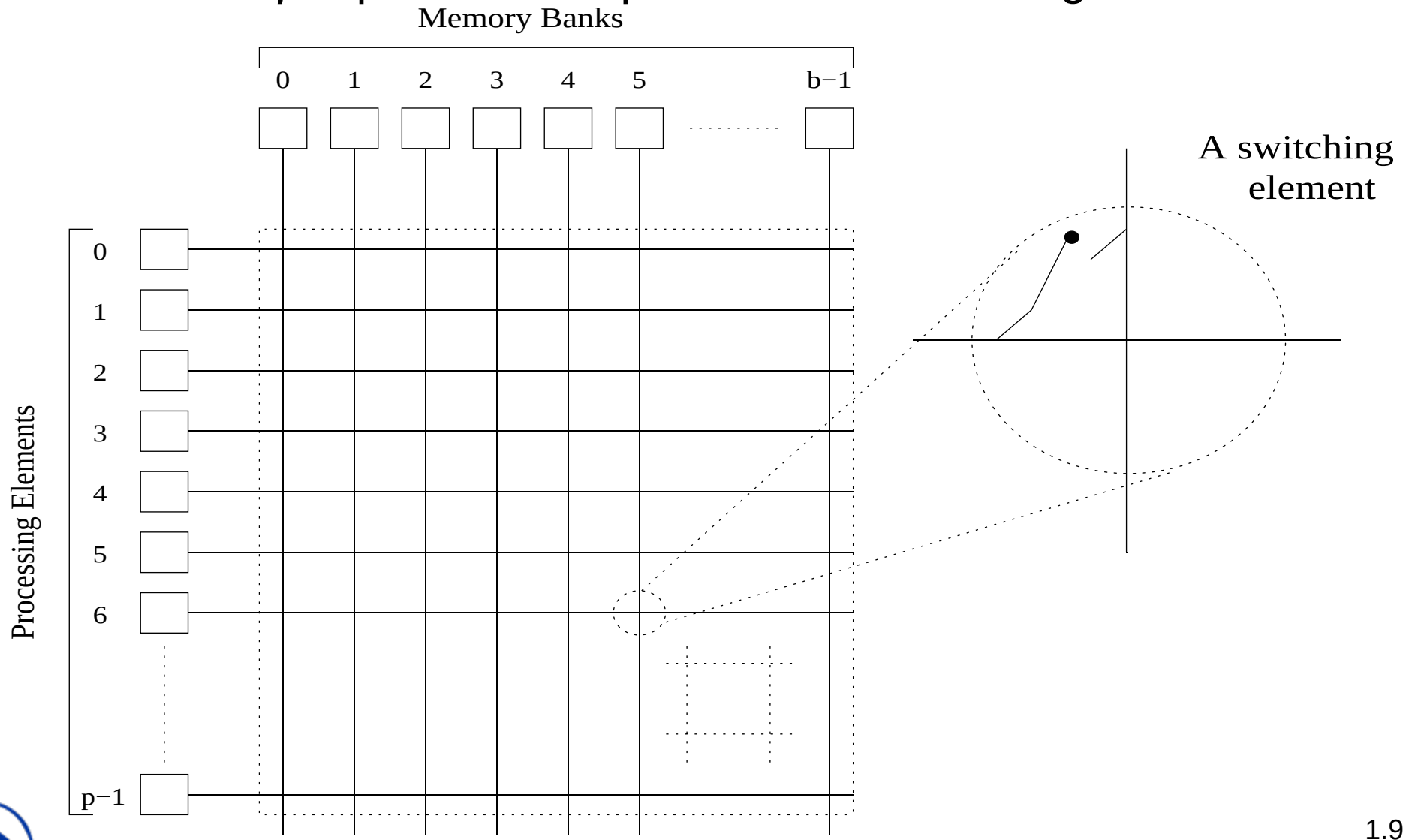
Bus-based interconnects (a) with no local caches; (b) with local memory/caches.

Since much of the data accessed by processors is local to the processor, a local memory can improve the performance of bus-based machines. Example??

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Network Topologies: Crossbars

A crossbar network uses an $p \times b$ grid of switches to connect p inputs to b outputs in a non-blocking manner.



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Network Topologies: Crossbars

- The cost of a crossbar of p processors grows as $O(p^2)$.
- This is generally difficult to scale for large values of p.
- Examples of machines that employ crossbars include the Sun Ultra HPC 10000 and the Fujitsu VPP500.



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Static Network Topologies

- Completely Connected
- Star
- Tree
- Linear Array & Ring
- 2-D mesh & Torus
- Hypercube
- k-d mesh/cube



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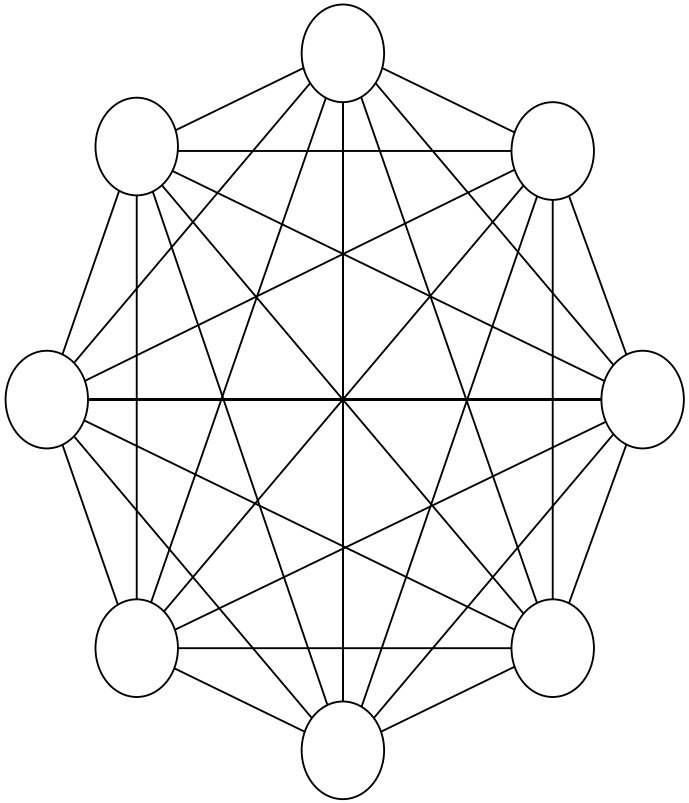
Network Topologies: Completely Connected Network

- Each processor is connected to every other processor.
- The number of links in the network scales as $O(p^2)$.
- While the performance scales very well, the hardware complexity is not realizable for large values of p.
- In this sense, these networks are static counterparts of crossbars.

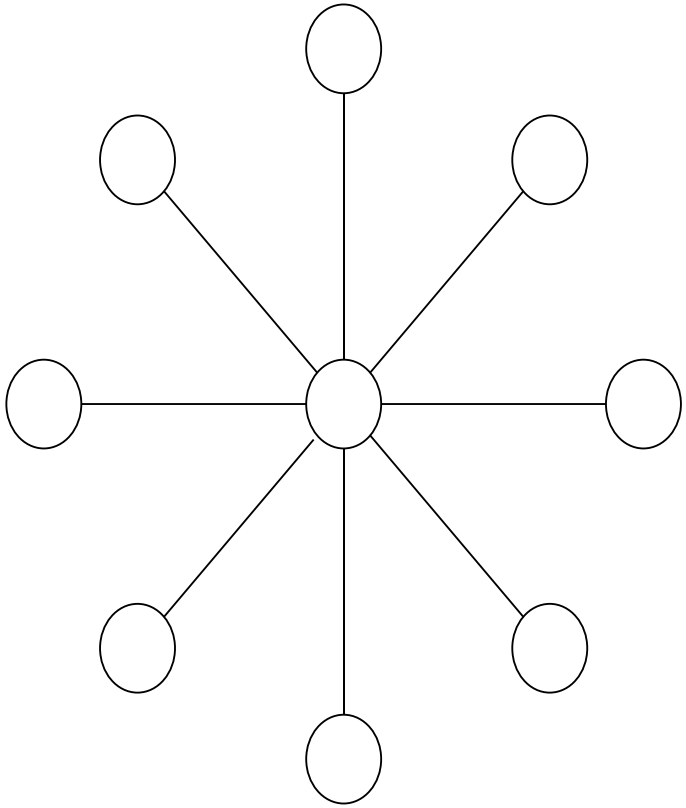


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Network Topologies: Completely Connected & Star Connected Networks



(a)



(b)

(a) A completely-connected network of eight nodes;
(b) a star connected network of nine nodes.

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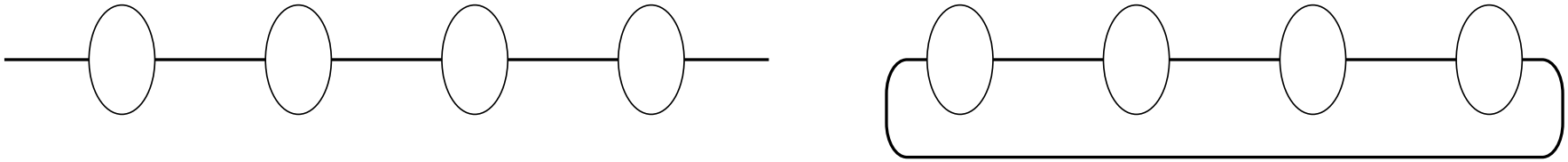
Network Topologies: Star Connected Network

- Every node is connected only to a common node at the center.
- Distance between any pair of nodes is 2. However, the central node becomes a bottleneck.
- In this sense, star connected networks are static counterparts of buses.



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Network Topologies: Linear Arrays

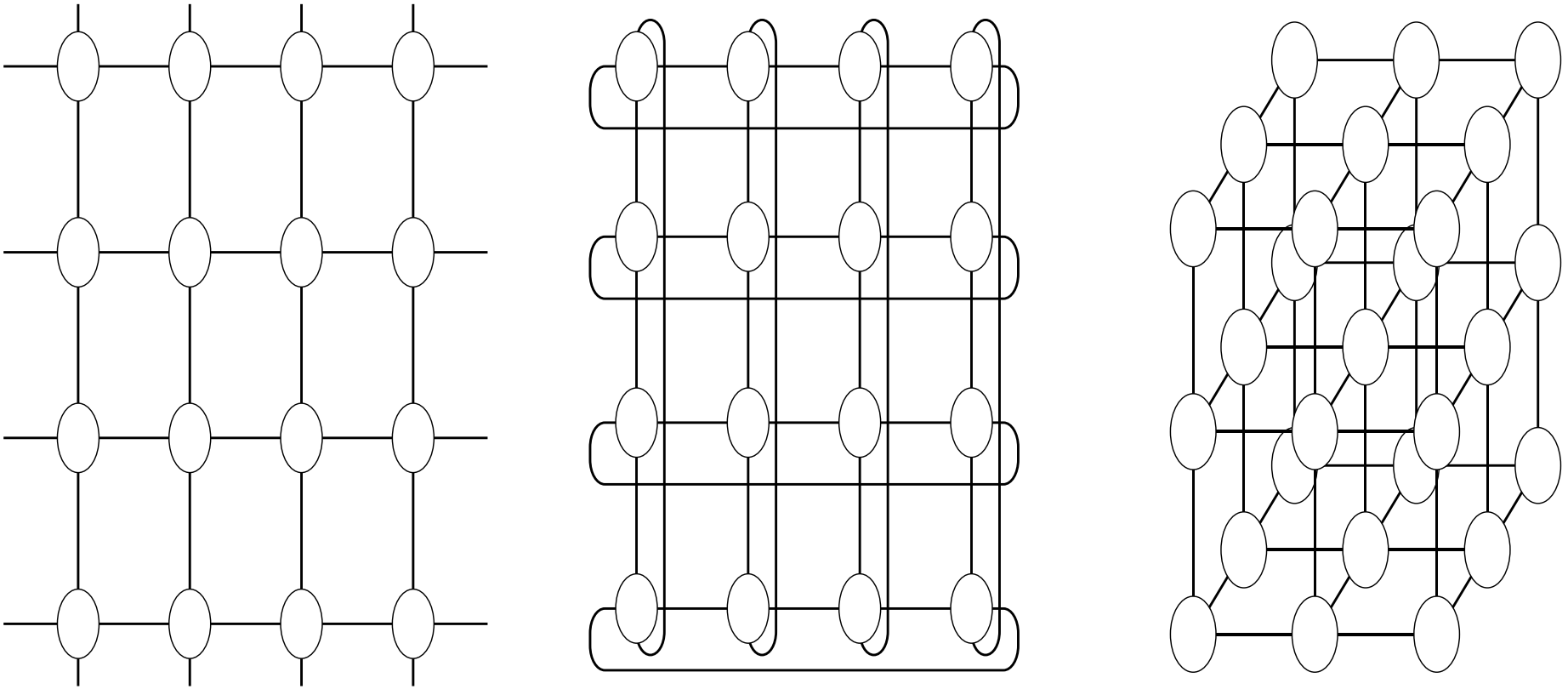


(a) (b)

Linear arrays: (a) with no wraparound links; (b) with wraparound link.

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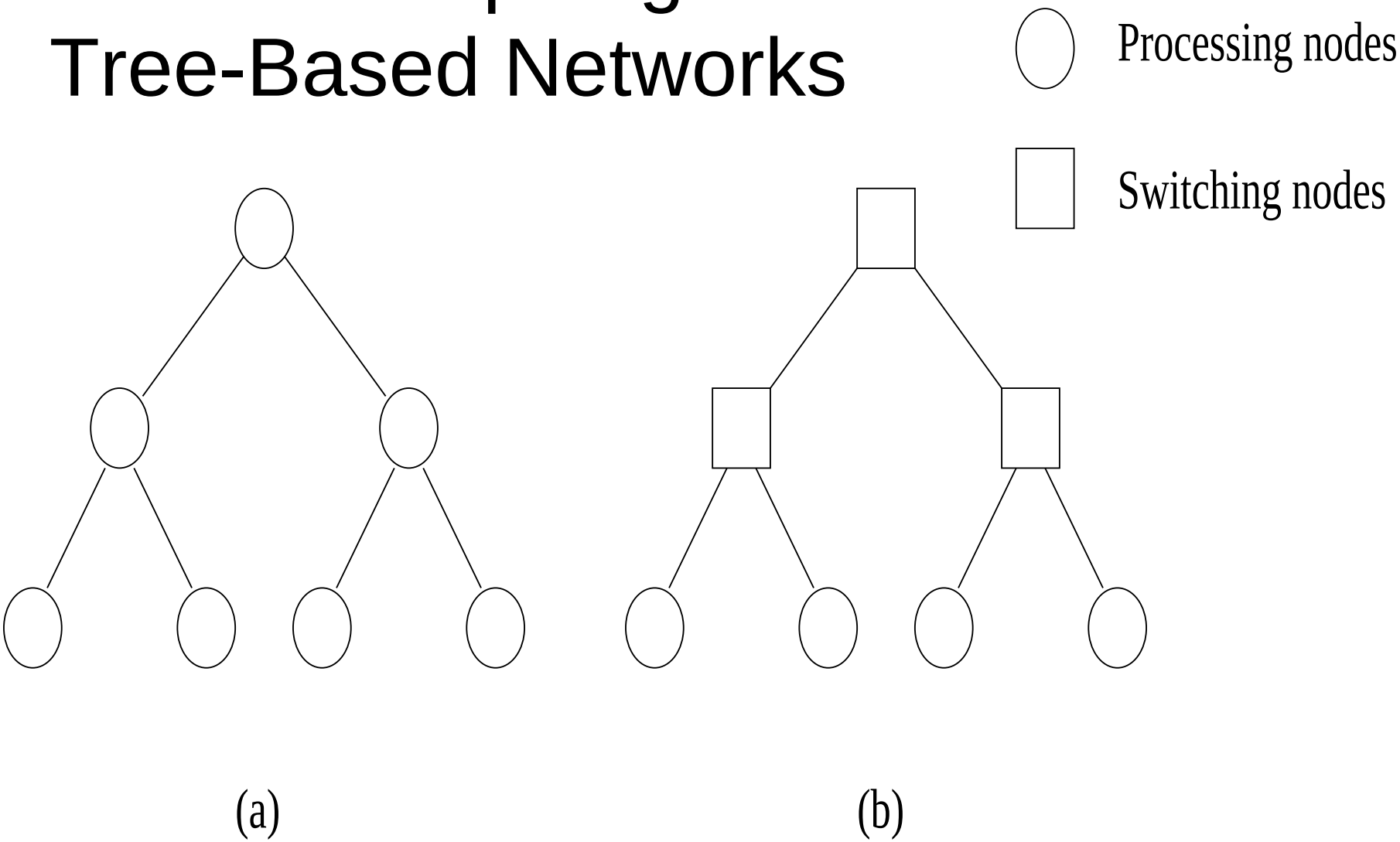
Network Topologies: Two- and Three Dimensional Meshes



Two and three dimensional meshes: (a) 2-D mesh with no wraparound; (b) 2-D mesh with wraparound link (2-D torus); and (c) a 3-D mesh with no wraparound.

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Network Topologies: Tree-Based Networks



Complete binary tree networks: (a) static tree network (b)dynamic tree network

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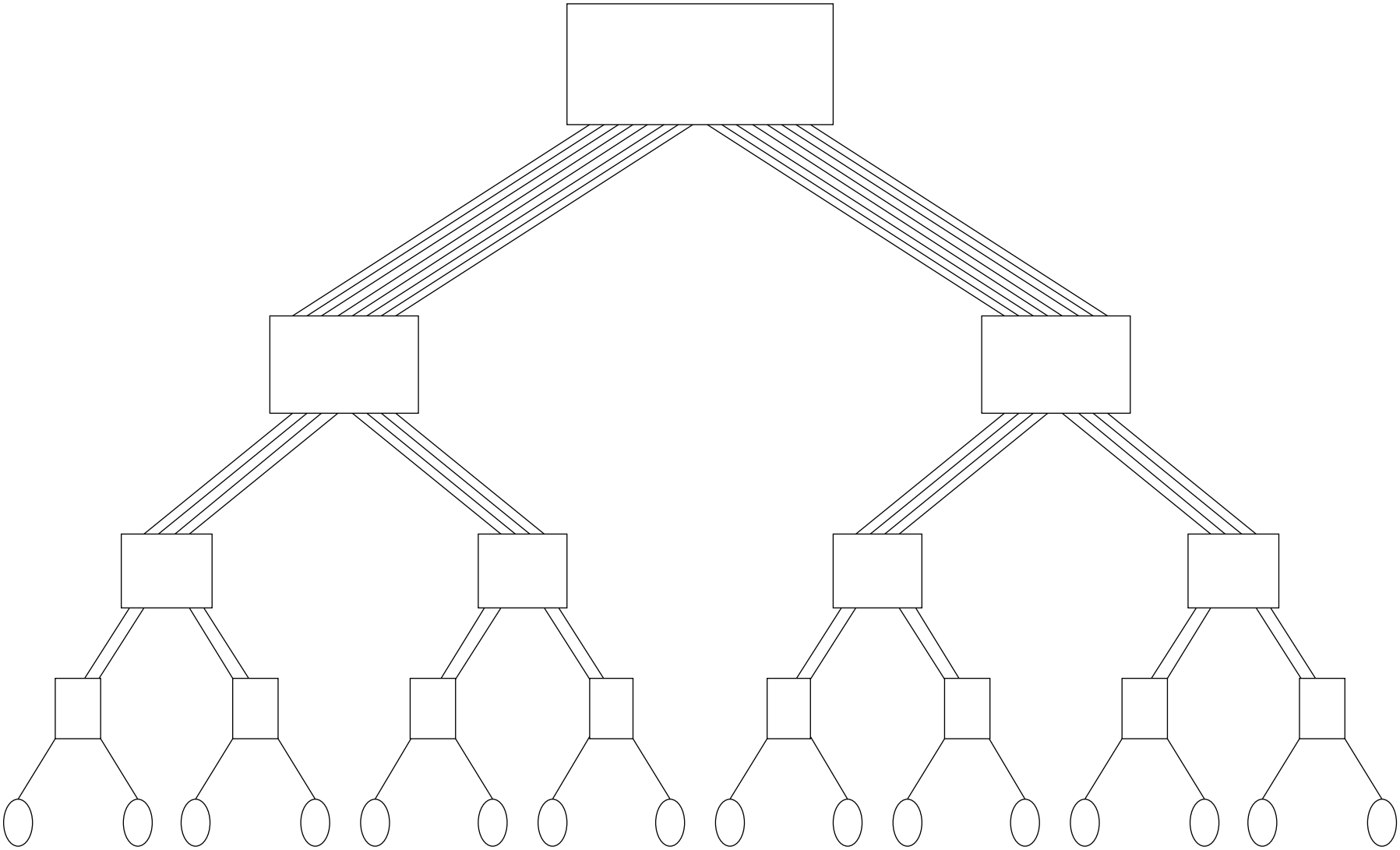
Network Topologies: Tree Properties

- The diameter is *2 [(logp+1)* -1] for p node complete binary tree.
- Links higher up the tree potentially carry more traffic than those at the lower levels.
- For this reason, a variant called a fat-tree, fattens the links as we go up the tree.
- Trees can be laid out in 2D with no wire crossings. This is an attractive property of trees.



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Network Topologies: Fat Trees

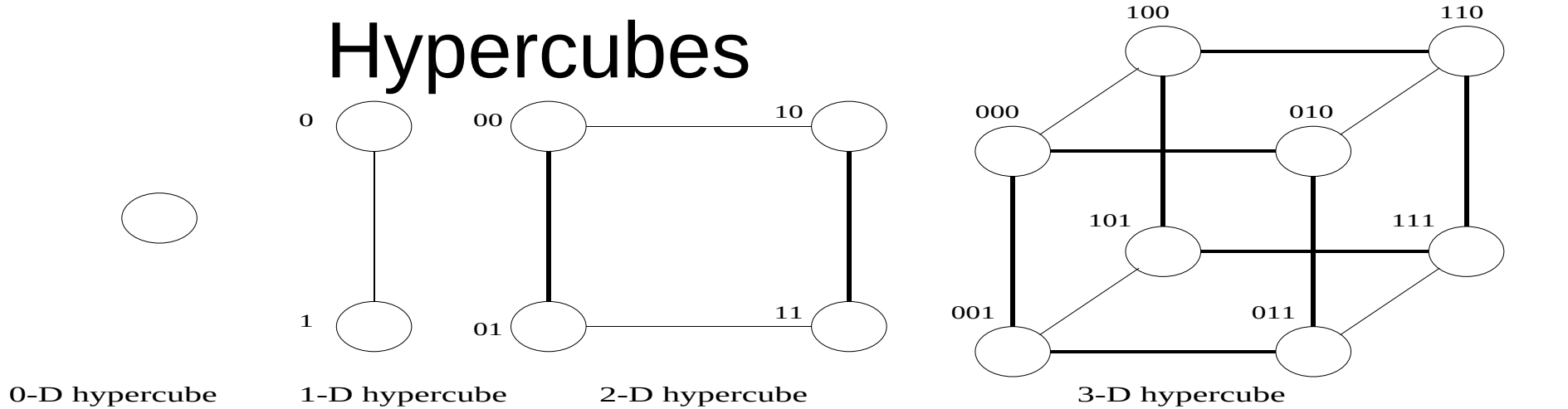


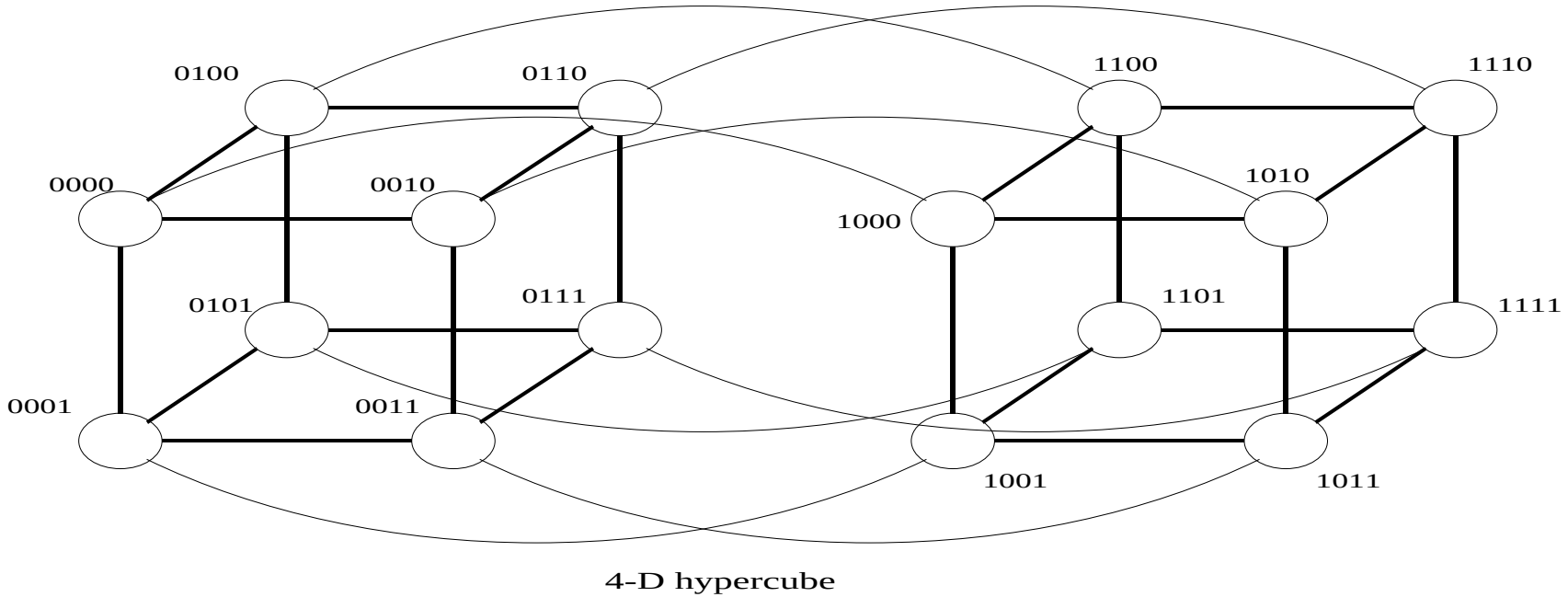
A fat tree network of 16 processing nodes.



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Network Topologies: Hypercubes





Construction of hypercubes from hypercubes of lower dimension.

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Network Topologies: Properties of Hypercubes

- The distance between any two nodes is at most *log p*.
- Each node has *log p* neighbors.
- The distance between two nodes is given by the number of bit positions at which the two nodes differ.
- Routing algorithm for Hypercube.



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